

1. What are Data Models and their Main Categories?

Data Models

- Set of concepts to describe database structure.
- Defines operations for manipulating data structures.
- Includes constraints that the database must obey.
- Provides a clear abstraction of data storage.

Conceptual (high-level) Models

- Concepts close to how users perceive data.
- Uses entities, attributes, and relationships.
- Independent of physical storage details.
- Example: Entity-Relationship (ER) model.

Physical (low-level) Models

- Describes how data is stored on disks.
- Defines record formats and byte orderings.
- Specifies access paths like primary indexes.
- Targeted at computer specialists and DBAs.

Implementation (representational) Models

- Concepts understood by end users easily.
- Hides some details of physical storage.
- Used by most commercial DBMS systems.
- Example: Relational, Network, and Hierarchical models.

2. What is the Difference between Database Schemas and Instances?

Database Schema

- Description of the overall database structure.
- Specified during the database design phase.

- Changes very infrequently over time.
- Also known as the database intension.

Database State (Instance)

- Actual data stored at a specific moment.
 - Captured when data is loaded or updated.
 - Changes every time a record is modified.
 - Also known as the database extension.
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3. How is the Three-Schema Architecture Structured?

Internal Level

- Describes physical storage structure of database.
- Uses a physical data model for details.
- Defines access paths for efficient searching.
- Lowest level of the database architecture.

Conceptual Level

- Describes structure for the community of users.
- Hides details of physical storage structures.
- Focuses on entities, data types, and constraints.
- Middle level that connects internal and external.

External Level (Views)

- Describes part of database for specific users.
 - Provides different views for different user groups.
 - Hides irrelevant data from unauthorized users.
 - Highest level closest to the end users.
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4. What is Data Independence?

Logical Data Independence

- Capacity to change the conceptual schema.
- Does not require changing external schemas.
- No need to rewrite application programs.
- Useful when adding new entities or attributes.

Physical Data Independence

- Capacity to change the internal schema.
 - Does not change conceptual or external schemas.
 - Useful for optimizing performance or storage.
 - Common when adding new indexes or files.
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5. What are the DBMS Languages and Interfaces?

DBMS Languages (DDL & DML)

- DDL defines the database conceptual schemas.
- DDL is used by DBA and designers.
- DML specifies data retrievals and updates.
- SQL is the standard language for relational.

DBMS Interfaces

- Menu-based interfaces for browsing and clicking.
 - Forms-based interfaces for entering specific data.
 - Graphical user interfaces (GUI) using diagrams.
 - Programmer APIs like JDBC for coding access.
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6. What are the Common DBMS Architectures?

Centralized Architecture

- Everything runs on a single computer system.
- Includes DBMS software and user display.
- All processing is done in one location.

- Traditional model for older legacy systems.

Client-Server Architecture

- Defined by specialized servers for data tasks.
- Clients provide the user interface processing.
- Servers provide database and print services.
- Connects multiple machines over a network.

Three-Tier Architecture

- Adds an Application Server between tiers.
- Handles business logic and data security.
- Client interacts only with the middle tier.
- Standard architecture for modern web apps.

7. How are Database Management Systems Classified?

By Data Model

- Relational models using tables and rows.
- Object-Oriented models using classes and objects.
- Legacy models like Network or Hierarchical.
- Modern NoSQL models for unstructured data.

By Number of Users and Site

- Single-user systems for personal desktop use.
- Multi-user systems for large organizations.
- Centralized systems located at one site.
- Distributed systems spread across many sites.

Chapter 5

1. What is the Relational Data Model?

Relational Model Concepts

- Represented as a collection of relations (tables).
- Each relation resembles a table of values.
- Data is stored as a set of rows (tuples).
- Each column header is called an attribute.

Relation Characteristics

- Ordering of tuples in a relation is not important.
 - Every value in a tuple is atomic (indivisible).
 - Attributes within a relation must have unique names.
 - Composite and multi-valued attributes are not allowed.
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2. What are the Different Types of Constraints?

Implicit and Explicit Constraints

- Inherent constraints are based on the data model.
- Schema-based constraints are explicitly defined in DDL.
- Domain constraints specify the valid set of values.
- Constraints must be enforced by the DBMS.

Key Constraints

- Superkey is a set of attributes identifying tuples.
 - Key is a minimal superkey with no redundancy.
 - Primary Key uniquely identifies rows in a table.
 - No two tuples can have the same key.
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3. How do Integrity Constraints Work?

Entity Integrity

- Primary key values cannot be NULL at all.

- This ensures every tuple is uniquely reachable.
- Enforced on every individual relation in database.
- Prevents records from being unidentified or lost.

Referential Integrity

- Involves relationship between two different database relations.
 - Foreign key must match a primary key value.
 - Ensures that a reference to data exists.
 - Prevents "dangling" references in the database system.
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4. What are Relational Update Operations?

Insert and Delete Operations

- Insert adds a new tuple to relation.
- Insert can violate any of the four constraints.
- Delete removes an existing tuple from relation.
- Delete can only violate referential integrity constraints.

Update Operation

- Changes values of attributes in existing tuples.
 - Updating a non-key attribute is usually safe.
 - Updating a key is like delete then insert.
 - DBMS must check all constraints after updates.
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5. How is the Relational Algebra Defined?

Unary Operators

- SELECT (σ) filters rows based on conditions.
- PROJECT (π) selects specific columns from table.
- RENAME (ρ) changes names of relation attributes.
- These operators work on a single relation.

Set Theory Operators

- UNION combines tuples from two compatible relations.

- INTERSECTION finds tuples common to both relations.
 - DIFFERENCE finds tuples in one but not other.
 - Relations must be "type compatible" for these.
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6. What are Join and Division Operations?

Join Operations

- Combines related tuples from two different relations.
- Theta Join uses a general comparison condition.
- Equijoin uses only the equality comparison operator.
- Natural Join removes duplicate attributes after joining.

Outer Joins and Division

- Left Outer Join keeps all rows from left.
 - Right Outer Join keeps all rows from right.
 - Full Outer Join keeps rows from both sides.
 - Division operator is used for "for all" queries.
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7. What are Aggregate Functions and Grouping?

Aggregate Functions

- Used to summarize data from many tuples.
- Common functions include SUM, AVERAGE, and COUNT.
- MAX and MIN find extreme values in data.
- Usually applied to a single attribute's values.

Grouping Concepts

- Tuples are partitioned based on grouping attributes.
- Aggregate functions are applied to each group.
- Result has one tuple for each group.
- Essential for generating reports and data analysis.